UNIVERSITY OF WAH (UOW)

DEPARTMENT OF COMPUTER SCIENCE

Project Report

Course Title: Artificial Intelligence

Course Code: (CS–202 L)

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1. Project Title:

HEADGEAR SENTINEL: AI-Based Personal Protective Equipment (PPE) Detection System

2. Abstract

Workplace safety is a critical concern in industrial, construction, and manufacturing environments where the absence of proper Personal Protective Equipment (PPE) can lead to severe injuries or fatalities. Manual monitoring of safety compliance is inefficient, time-consuming, and prone to human error. To address this issue, this project presents **HeadGear Sentinel**, an AI-based image detection system designed to automatically identify the presence or absence of safety equipment in images.

The proposed system uses a deep learning-based object detection approach to detect PPE items such as helmets, gloves, safety vests, boots, and humans from images. The system is trained **from scratch** using the YOLOv8 object detection architecture and a labeled PPE dataset in YOLO format. A clean and user-friendly desktop application is developed using **Python and Tkinter**, allowing users to train the model, perform image-based detection, and view detailed results within a single interface.

The system supports real model training, single image and folder-based detection, visual bounding box outputs, confidence scores, and safety percentage analysis. Experimental results demonstrate that the model can effectively identify safety compliance and missing equipment under various conditions. HeadGear Sentinel offers a practical, scalable, and cost-effective solution for automated safety monitoring and has strong potential for real-world deployment in industrial environments.

3. Introduction

Object detection is a fundamental problem in computer vision that involves identifying and localizing objects within an image. With advancements in deep learning, object detection has become increasingly accurate and efficient, enabling applications in surveillance, healthcare, autonomous systems, and industrial automation.

In safety-critical environments such as construction sites and factories, ensuring the use of Personal Protective Equipment (PPE) is essential. Traditional monitoring methods rely on manual supervision, which is unreliable and difficult to scale. This project is motivated by the need for an automated system that can assist in enforcing safety compliance using artificial intelligence.

HeadGear Sentinel aims to leverage modern object detection techniques to provide an intelligent, image-based PPE detection system that is easy to use, accurate, and suitable for academic as well as real-world use.

4. Problem Definition

The problem addressed in this project is the lack of an automated, reliable, and scalable solution for monitoring PPE compliance in images. Existing manual systems are inefficient and error-prone.

Project Boundaries:

- Image-based detection only (no video or real-time streaming)
- Desktop application environment
- YOLO-formatted dataset

Assumptions:

- Input images are reasonably clear
- Dataset annotations are correct

Constraints:

- Limited computational resources
- Training performed without pretrained weights

5. Literature Review

Several object detection models such as R-CNN, SSD, Faster R-CNN, and YOLO have been proposed in recent years. Among these, YOLO (You Only Look Once) is widely recognized for its speed and accuracy. Previous studies have applied YOLO-based models for PPE detection, often relying on pretrained weights.

However, many existing systems focus on real-time video processing and lack a structured GUI or academic transparency. Additionally, limited work has been done on training PPE detection models entirely from scratch for educational purposes. This project addresses these gaps by implementing a clean, explainable system with real training and a graphical interface.

6. System Overview

The system consists of three main components:

1. **Training Module** – Handles dataset validation and model training
2. **Detection Module** – Performs object detection on images
3. **Results Module** – Displays detection summaries, safety percentage, and visual outputs

Data Flow:
Dataset → Training → Trained Model → Image Detection → Results Visualization

7. Dataset Description

The dataset used in this project is a labeled PPE image dataset obtained from a public source. It follows the YOLO directory structure with separate folders for training, validation, and testing images and labels.

Classes Include:

- Person
- Helmet
- Mask
- Gloves
- Safety Vest
- Boots
- Goggles

Preprocessing Steps:

- Image resizing to 640×640
- Normalization
- Label verification

Annotations are provided in YOLO format using bounding boxes.

8. Methodology

The YOLOv8 object detection algorithm is used due to its efficiency and accuracy. The model is initialized using the YOLOv8 architecture configuration file and trained from scratch using the labeled dataset.

The training pipeline includes:

- Data loading
- Forward propagation
- Loss computation (classification, localization, confidence)
- Backpropagation
- Optimization using gradient descent

Model performance is evaluated using detection confidence and accuracy trends.

9. Implementation Details

Programming Language: Python
Frameworks & Libraries:

- Ultralytics YOLOv8
- Tkinter
- OpenCV
- NumPy
- Pillow

Hardware:

- Google Colab GPU (for training)
- Local system (for GUI & detection)

Implementation Decisions:

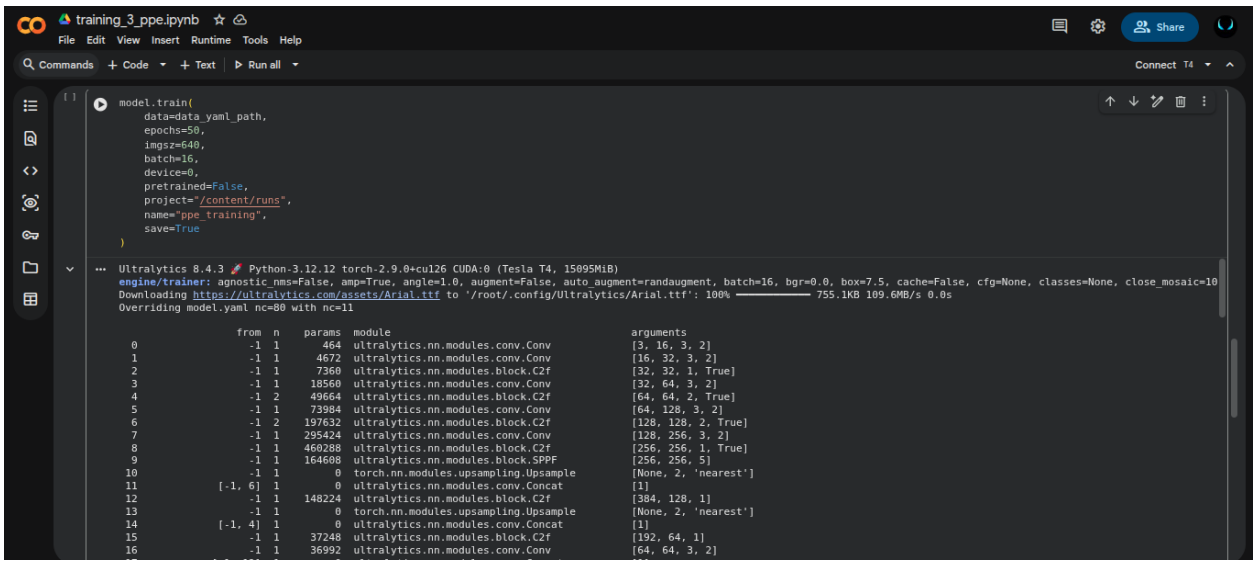
- Single Python file design
- Modular GUI layout
- Real training without pretrained weights

10. Codes and Output Screenshots:

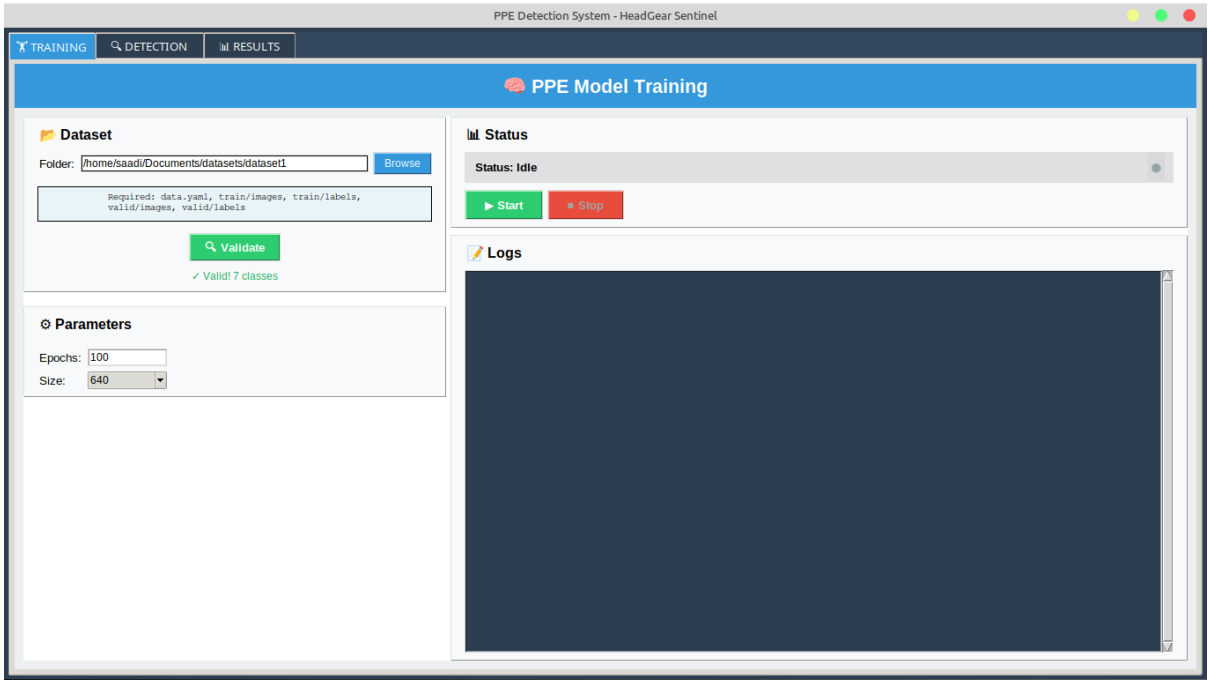
The training code:

```
def run_training(self, epochs, img_size):
"""Run training"""
try:
self.update_status("Training...", '#f39c12')
self.log_message("Starting YOLOv8 training...")
self.log_message(f"Epochs={epochs}, Size={img_size}")
model = YOLO("yolov8n.yaml")
if os.path.exists(self.trained_model_path):
self.log_message("Resuming from checkpoint...")
model = YOLO(self.trained_model_path)
yaml_path = os.path.join(self.dataset_path, 'data.yaml')
results = model.train(
data=yaml_path,
epochs=epochs,
imgsz=img_size,
pretrained=False,
resume=os.path.exists(self.trained_model_path),
project=".",
name="runs",
save=True,
exist_ok=True
)
model.save(self.trained_model_path)
self.log_message("✓ Training complete!")
```

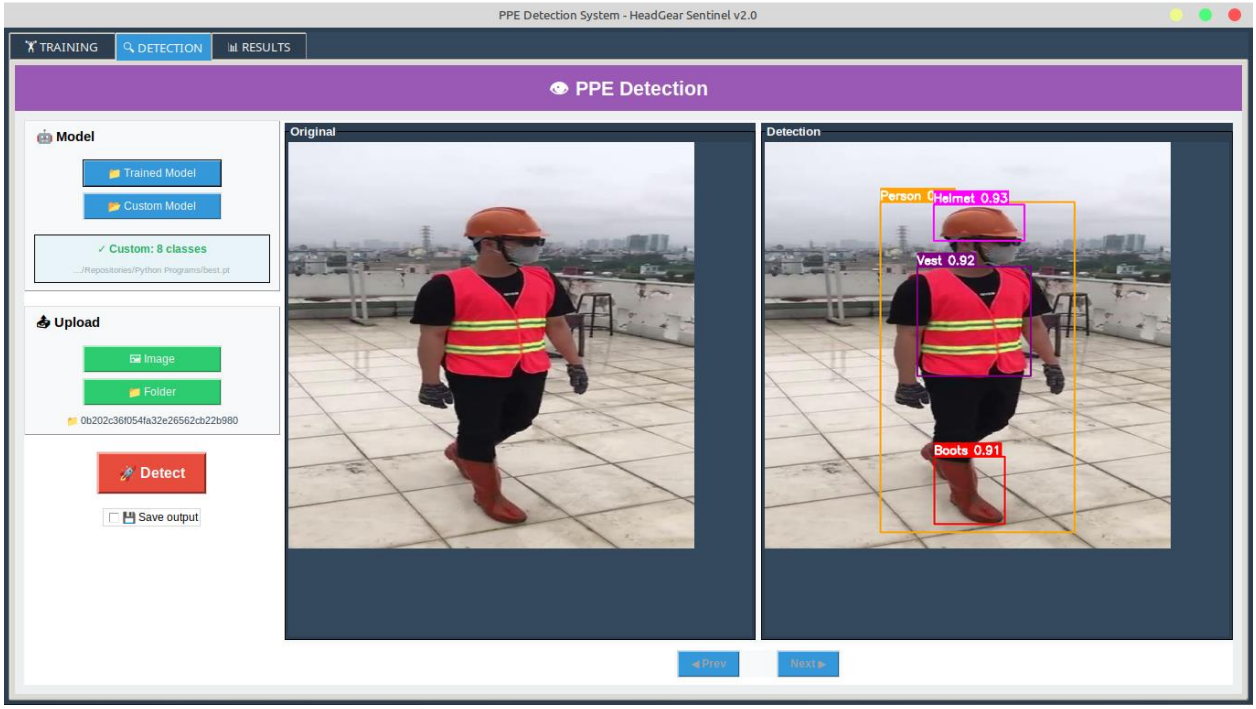
Training logs(Google Colab):



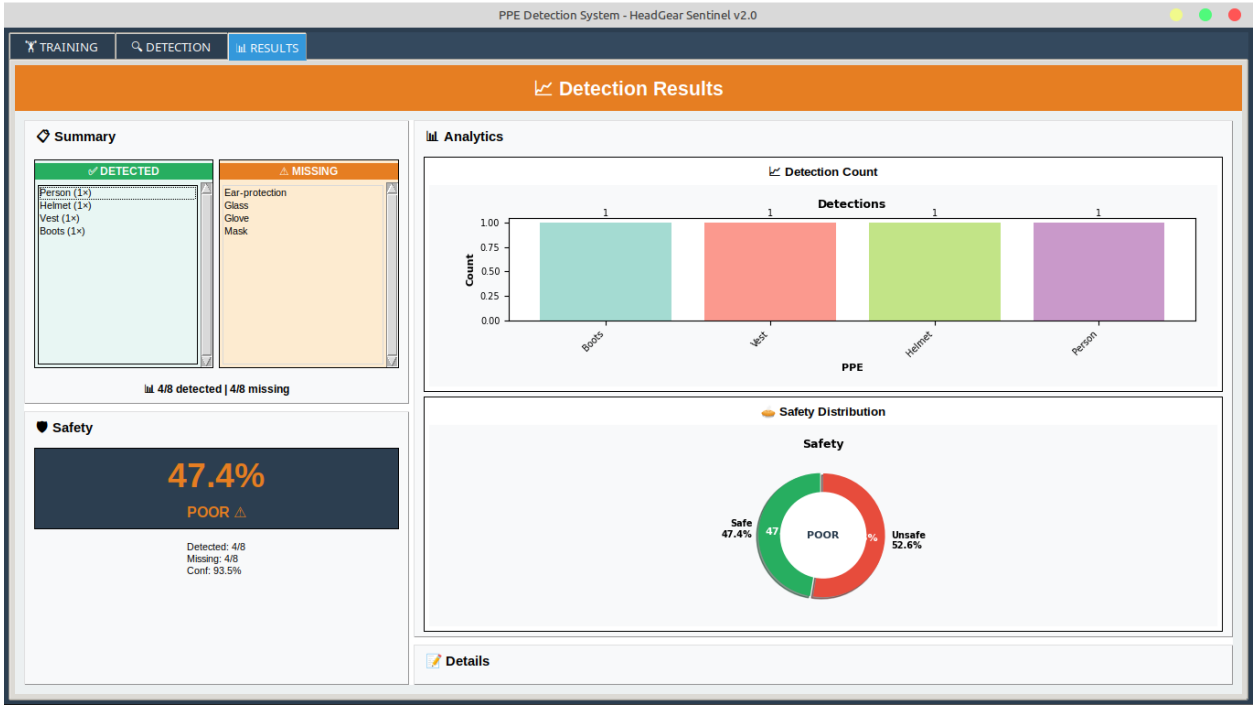
Training window:



Detection window:



Results window:



11. Results and Analysis

The HeadGear Sentinel system was evaluated on test images containing various Personal Protective Equipment (PPE) scenarios. The trained YOLOv8 model successfully detected

safety equipment and identified missing items with visible bounding boxes and confidence scores. A safety compliance percentage was calculated based on detected PPE per person.

Key Observations:

- Accurate detection under normal lighting conditions
- Clear visualization of detected and missing PPE
- Stable learning behavior observed through training graphs

12. Discussion

The experimental results indicate that training an object detection model from scratch is practical for PPE detection when sufficient data quality and training epochs are provided. Although the absence of pretrained weights increased training time, it ensured complete academic transparency and better understanding of the learning process. The integration of a Tkinter-based graphical interface enhanced usability and made the system more accessible for non-technical users.

13. Challenges and Limitations

The development and training process involved several challenges, including:

- Long training duration due to training from scratch
- Limited GPU availability and hardware constraints
- Sensitivity to low-quality or poorly lit images
- Restricted dataset diversity affecting generalization

These limitations may impact performance in complex real-world environments.

14. Ethical Considerations

Ethical responsibility was considered throughout the project. The system uses publicly available datasets and does not involve facial recognition or personal identity tracking. Possible bias due to dataset imbalance has been acknowledged. The application is designed strictly for safety compliance and educational use, avoiding misuse for invasive surveillance.

15. Applications and Use Cases

The developed system can be applied in multiple real-world and academic scenarios, including:

- Construction site safety audits
- Industrial and factory compliance monitoring
- Safety training and awareness programs
- Academic demonstrations of AI-based object detection

16. Conclusion

HeadGear Sentinel successfully demonstrates an AI-based PPE detection system developed with real model training and a complete desktop application. The project meets its objectives by combining object detection, safety analysis, and an intuitive user interface. The results confirm the effectiveness of deep learning techniques for automated safety monitoring in controlled environments.

17. Future Work

Several enhancements can be considered to extend the system's capabilities:

- Real-time video-based PPE detection
- Improved accuracy through dataset expansion
- Model optimization for faster inference
- Cloud-based deployment and mobile integration

18. References

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